

Reg. No.: _____

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Amrita School of Computing, Bengaluru

B.Tech Missed Mid Semester Examination – October 2025

Seventh Semester CSE

Computer Science and Engineering

19CSE353 Mining of Massive Datasets

Duration : Two hours

Maximum : 50 Marks

CO	Course Outcomes
CO1	Understand the importance of data to business decisions, strategy and behavior. Predictive analytics, data mining and machine learning as tools give new methods for analyzing massive data sets.
CO2	Expose the students to big data systems like Hadoop, Spark and Hive.
CO3	Explore means to deal with huge document databases and infinite streams of data to mining large social networks and web graphs. Also, learn Algorithms suitable for large scale mining.
CO4	Use of analytic tools, case-studies to provide firsthand insight into how big-data problems and their solutions are commercially derived.
CO5	Design Large Scale Machine Learning algorithms with Practical hands-on experience for analyzing very large amounts of data.

Answer all the questions

1. You have four sets of integers representing different user activities, and you want to compare these sets to determine their similarity using MinHash signatures.

Set X: {1, 2, 3, 4, 5}, Set Y: {3, 4, 5, 6, 7}, Set Z: {1, 5, 8, 9}, Set W: {2, 5, 10, 11}.

Three hash functions used are

$$h_1(x) = (2x + 3) \bmod 11$$

$$h_2(x) = (x + 7) \bmod 13$$

$$h_3(x) = (3x - 5) \bmod 17$$

- i. Find the similarity of these sets and verify it using the Jaccard Similarity measure. [6] [CO3] [BTL3]
 - ii. Compute the L_1 -norm and L_∞ -norm for the sets X and Y. [2] [CO3] [BTL3]
 - iii. Compute the Cosine distance between the sets Z and W. [2] [CO3] [BTL3]
2. a. Explain the roles of the Map and Reduce tasks in the MapReduce model. [2+2] [CO2] [BTL1]
Using the word count example, illustrate how key-value pairs are generated and processed at each phase.

- b. Describe the process of grouping by key and the function of combiners in MapReduce. [2+1] [CO2]
[BTL1]
How do combiners improve the performance of a MapReduce job?
- c. How are failures of the Mapper, Reducer, and Master tasks handled during a MapReduce job? [3] [CO2]
[BTL1]
3. Calculate the PageRank values for each page in Fig. 1 using two iterations of the PageRank algorithm, assuming a damping factor of 0.85. Explain how the damping factor affects the PageRank results. [4+1] [CO3]
[BTL3]

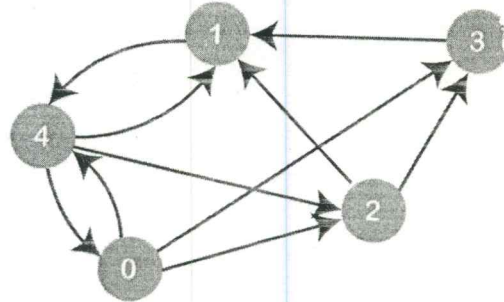


Fig. 1

4. The database given below in Table 1 has five transactions from five customers. Using the PCY algorithm, find the association rules in this set, if the minimum support (coverage) is 60% and the minimum confidence (accuracy) is 80%. [Note: minimum support threshold is 60% of the total transactions] [5] [CO3]
[BTL3]

Table 1

Customer	Items Purchased
1	Tomato, Potato, Onion
2	Onion, Carrot, Tomato
3	Tomato, Potato
4	Tomato, Potato, Spinach
5	Onion, Potato

5. Imagine you are evaluating the trustworthiness of several web pages using TrustRank. Two seed sites, a trusted news portal (S1) and a reputable university website (S2), are assigned the maximum TrustRank score of 1.0. Other pages include P1, a technology blog; P2, a product review site; P3, an affiliate marketing page; and P4, a spammy link farm. The link structure is as follows: S1 links to P1 and P2, S2 links to P2, P1 links to P3, P2 links to P3 and P4, and P3 links to P4. P4 also receives numerous links from low-quality or irrelevant external sites. Assume each page distributes its TrustRank evenly across its outgoing links and that you propagate trust for two iterations. [2+2+1] [CO3]
[BTL3]

- i. Calculate the TrustRank scores for pages P1, P2, P3, and P4 after two iterations, showing intermediate steps.
 - ii. Based on the computed TrustRank values, identify which page is most likely to be spam and justify your answer briefly.
 - iii. If P4 has a high PageRank but the lowest TrustRank, explain what this reveals about the quality of its inbound links.

6. Suppose that the task is to cluster points (with (x, y) representing location) into two clusters, where the points are $A_1(1,9)$, $A_2(2,5)$, $A_3(8,4)$, $B_1(5,8)$, $B_2(7,5)$, $B_3(6,4)$. The distance function is Euclidean distance. Suppose initially A_1 and B_1 are the center of each cluster, respectively. [3+1+1] [CO3] [BTL3]
 - i. Use the k-means algorithm to show the final two clusters or two more iterations, whichever occurs first.
 - ii. Briefly explain how the clustering might differ if the CURE algorithm were applied instead.
 - iii. Highlight at least one advantage of CURE over k-means in handling such points.

7. You are given the following 2-dimensional dataset of 10 points: [5] [CO3] [BTL4]
 $\{(2,3), (3,4), (5,6), (10,12), (11,14), (9,10), (15,16), (16,18), (18,20), (1,1)\}$
 You are required to cluster these points using the BFR algorithm. The number of clusters is $k = 3$, and your memory can hold only 6 points at a time.

8. Consider a small social network graph given in Fig. 2 with nodes including P, Q, and R. Suppose there is an edge between P and Q, and an edge between Q and R. [4+1] [CO4] [BTL4]
 - i. What is the probability that an edge also exists between P and R?
 - ii. Based on your calculation, does this network exhibit locality of relationships (i.e., friends of friends are likely to also be friends), as is typical in social networks?

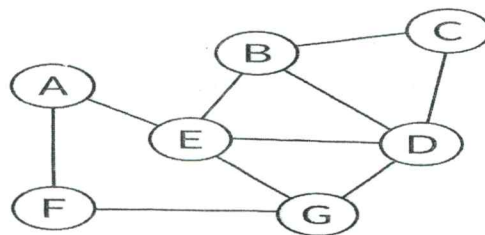


Fig. 2

Course Outcome / Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO1	-	BTL 1	10
CO2	10	BTL 2	-
CO3	35	BTL 3	30
CO4	5	BTL 4	10
CO5	-	BTL 5	-
		BTL 6	-